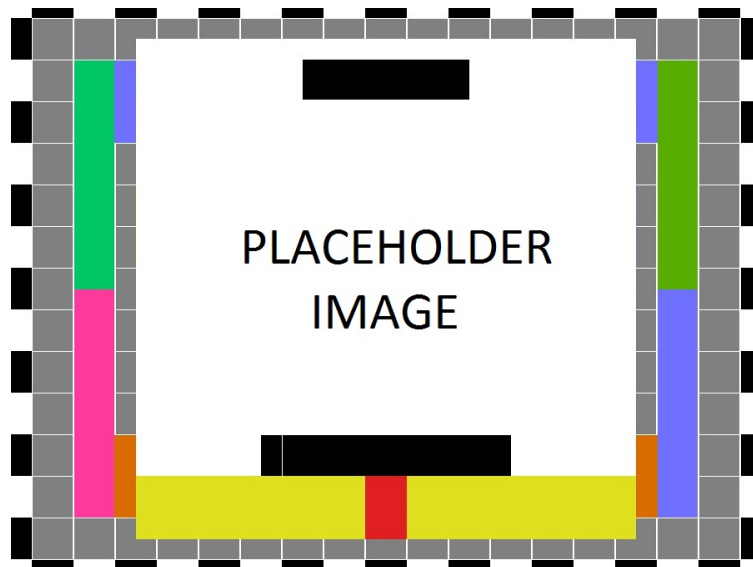


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**SYSTEM REQUIREMENTS SPECIFICATION
CSE 4316: SENIOR DESIGN I
FALL 2026**



**TEAM TRAILBLAZER
TERRASCOUT AUTONOMOUS GROUND ROVER**

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REVISION HISTORY

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1 INTRODUCTION

This SRS specifies the requirements for TerraScout Autonomous Ground Rover, an autonomous ground rover for photovoltaic (PV) field inspection. It follows the SRS information item of ISO/IEC/IEEE 29148:2018 [1].

1.1 PURPOSE

This document is the agreed requirements baseline between Team Trailblazer and the sponsor. Requirements herein may not be changed without the sponsor's agreement.

1.2 SCOPE

The system shall autonomously patrol drivable lanes of a PV field along operator-defined GPS waypoints, capture RGB and thermal imagery of panels, detect likely faults on-board, produce a GPS-tagged inspection report, and return to a charging dock autonomously. Off-board analytics dashboards and cloud storage are out of scope for the prototype.

1.3 PRODUCT OVERVIEW

1.3.1 PRODUCT PERSPECTIVE

The rover is a self-contained mobile system that interoperates with two external elements: an operator ground-station laptop (over Wi-Fi) and a passive charging dock. See Figure 1.

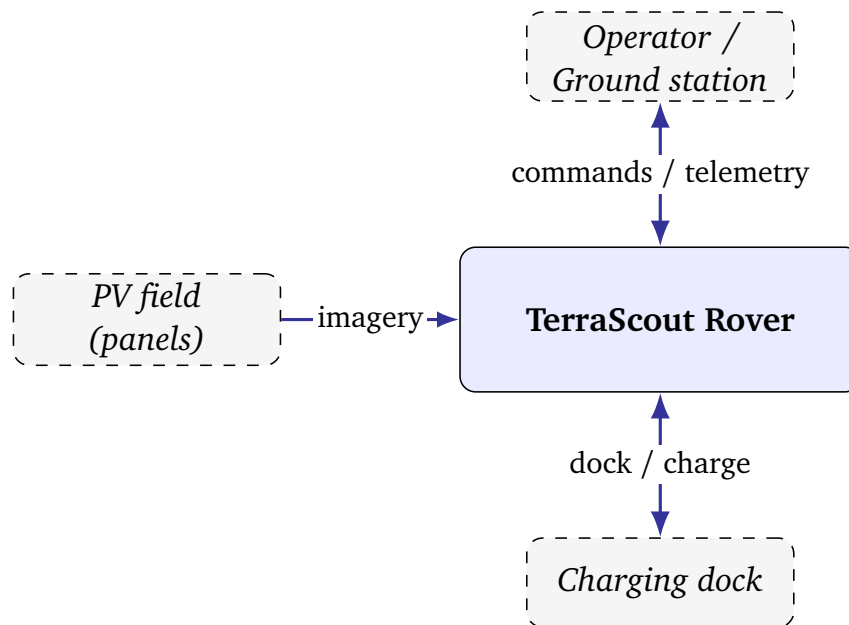


Figure 1: System context: rover, ground station, and charging dock

1.3.2 PRODUCT FUNCTIONS

Tele-operated and autonomous driving; waypoint navigation with obstacle avoidance; RGB/thermal capture; on-board fault detection; GPS-tagged report generation; autonomous dock and recharge; ground-station monitoring and control.

1.3.3 USER CHARACTERISTICS

The primary user is a field O&M technician: comfortable with tablets/laptops, not a roboticist, operating outdoors in PPE. A secondary maintainer role (team/engineer) services hardware and software.

1.3.4 ASSUMPTIONS & DEPENDENCIES

Drivable lanes $\leq 10\%$ grade; RTK-GPS correction available; private Wi-Fi between rover and ground station; daylight, dry-weather operation.

1.4 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

Term	Definition
PV	Photovoltaic (solar) panel
Hotspot	Localized thermal anomaly indicating a likely panel fault
RTK	Real-Time Kinematic GPS correction (centimeter-level positioning)
Waypoint	An operator-defined GPS coordinate on the patrol route
Ground station	Operator laptop running the control/monitoring web page

2 REQUIREMENTS ENGINEERING APPROACH

Requirements use identifiers [TYPE] - [NN] (EI, FR, US, PE, PK, SA, SE, MS, QA) and carry the ISO/IEC/IEEE 29148 attribute set: Description, Rationale, Source, Priority (Critical/High/Moderate/Low/Future), Verification Method (Inspection/Analysis/Demonstration/Test), Constraints, and Applicable Standards. Each requirement is written to be singular, unambiguous, and verifiable.

3 EXTERNAL INTERFACE REQUIREMENTS

3.1 EI-01: GROUND-STATION CONTROL LINK

DESCRIPTION

The rover shall expose a WebSocket control/telemetry link over TLS on the private Wi-Fi network, accepting start/stop/return commands and streaming pose, battery, and status at ≥ 2 Hz.

RATIONALE

The operator must command runs and observe live status from the ground station.

SOURCE

Sponsor scoping interview; team.

PRIORITY

Critical

VERIFICATION METHOD

Test — exercise each command and measure telemetry rate.

CONSTRAINTS

Wi-Fi range limits operating radius; link loss triggers safe-stop (see FR-05).

APPLICABLE STANDARDS

IETF RFC 6455 (WebSocket), RFC 8446 (TLS 1.3).

3.2 EI-02: CHARGING DOCK INTERFACE

DESCRIPTION

The rover shall mate with the charging dock and draw charge current through a polarized contact pair when docked.

RATIONALE

Enables autonomous recharge without operator handling.

SOURCE

Team.

PRIORITY

High

VERIFICATION METHOD

Demonstration — rover docks and charging begins.

CONSTRAINTS

Contact alignment tolerance ± 20 mm from approach heading.

APPLICABLE STANDARDS

UL 2089 (vehicle battery charging), as guidance.

4 FUNCTIONAL REQUIREMENTS

4.1 FR-01: WAYPOINT PATROL

DESCRIPTION

Given an ordered list of GPS waypoints, the rover shall drive to each in sequence within a 0.5 m cross-track tolerance and signal completion.

RATIONALE

Consistent, repeatable coverage of PV rows is the core inspection function.

SOURCE

Sponsor.

PRIORITY

Critical

VERIFICATION METHOD

Test — run a surveyed course; measure cross-track error.

CONSTRAINTS

Requires RTK correction; degraded to 2 m tolerance without RTK.

APPLICABLE STANDARDS

None specific.

4.2 FR-02: IMAGERY CAPTURE

DESCRIPTION

At each panel-row segment the rover shall capture time- and GPS-stamped RGB and thermal frames covering the adjacent panel faces.

RATIONALE

Imagery is the evidence from which faults are detected and reported.

SOURCE

Sponsor.

PRIORITY

Critical

VERIFICATION METHOD

Test — verify frames exist, are stamped, and cover the panels.

CONSTRAINTS

Thermal resolution limited by loaned camera.

APPLICABLE STANDARDS

EXIF GPS tagging.

4.3 FR-03: ON-BOARD FAULT DETECTION

DESCRIPTION

The rover shall classify each captured panel region as *nominal* or *suspected fault* (hotspot/soiling/crack) and record the location.

RATIONALE

Detecting faults on-board avoids shipping all imagery for manual review.

SOURCE

Sponsor; team.

PRIORITY

High

VERIFICATION METHOD

Test — run against a labeled validation set; report precision/recall (see PE-03).

CONSTRAINTS

Model runs within the compute budget of the on-board module.

APPLICABLE STANDARDS

None specific.

4.4 FR-04: INSPECTION REPORT

DESCRIPTION

At end of run the rover shall produce a report listing each suspected fault with GPS coordinate, timestamp, class, and a reference image, downloadable from the ground station.

RATIONALE

The report is the deliverable the O&M team acts on.

SOURCE

Sponsor.

PRIORITY

Critical

VERIFICATION METHOD

Demonstration — generate and download a report after a run.

CONSTRAINTS

Report format agreed with sponsor (CSV + image bundle).

APPLICABLE STANDARDS

None specific.

4.5 FR-05: SAFE-STOP & AUTONOMOUS RETURN

DESCRIPTION

On obstacle, low battery ($\leq 20\%$), or control-link loss > 10 s, the rover shall halt safely and, when able, navigate autonomously to the charging dock.

RATIONALE

Protects the platform and surroundings and enables unattended operation.

SOURCE

Team; safety review.

PRIORITY

Critical

VERIFICATION METHOD

Test — inject each trigger and observe safe-stop/return.

CONSTRAINTS

Return requires a known dock waypoint and sufficient reserve charge.

APPLICABLE STANDARDS

None specific.

5 USABILITY REQUIREMENTS

5.1 US-01: ONE-PAGE OPERATION

DESCRIPTION

A trained technician shall start a patrol, monitor it, and download the report using a single ground-station web page, completing setup in under 3 minutes in ≥ 8 of 10 trials.

RATIONALE

Field staff are not roboticists; operation must be simple.

SOURCE

Sponsor.

PRIORITY

High

VERIFICATION METHOD

Test — moderated usability trials.

CONSTRAINTS

Readable outdoors in daylight on a laptop screen.

APPLICABLE STANDARDS

WCAG 2.2 AA for the web page.

6 PERFORMANCE REQUIREMENTS

6.1 PE-01: PATROL THROUGHPUT

DESCRIPTION

The rover shall inspect at least 200 panels per hour under nominal conditions.

RATIONALE

Determines how much of a site can be covered per charge/day.

SOURCE

Sponsor.

PRIORITY

High

VERIFICATION METHOD

Test — timed field run.

CONSTRAINTS

Bounded by drive speed and capture dwell time.

APPLICABLE STANDARDS

None.

6.2 PE-02: ENDURANCE

DESCRIPTION

The rover shall operate for at least 45 minutes of continuous patrol on one charge.

RATIONALE

Sets usable run length between docks.

SOURCE

Team.

PRIORITY

High

VERIFICATION METHOD

Test — discharge run to 20%.

CONSTRAINTS

Battery capacity/weight budget.

APPLICABLE STANDARDS

None.

6.3 PE-03: DETECTION QUALITY

DESCRIPTION

On-board fault detection shall achieve ≥ 0.80 recall and ≥ 0.70 precision on the sponsor-provided validation set.

RATIONALE

Missed faults defeat the purpose; excessive false alarms waste truck-rolls.

SOURCE

Sponsor; team.

PRIORITY

High

VERIFICATION METHOD

Analysis — evaluate against labeled set.

CONSTRAINTS

Limited by training data volume.

APPLICABLE STANDARDS

None.

7 PACKAGING & DEPLOYMENT REQUIREMENTS

7.1 PK-01: FIELD-DEPLOYABLE UNIT

DESCRIPTION

The rover shall be delivered as a single assembled unit with the charging dock and a USB drive containing the ground-station application and operator manual.

RATIONALE

The sponsor receives a ready-to-run system, not a kit.

SOURCE

Sponsor.

PRIORITY

Moderate

VERIFICATION METHOD

Inspection.

CONSTRAINTS

Two-person liftable (<25 kg).

APPLICABLE STANDARDS

None.

8 SAFETY REQUIREMENTS

8.1 SA-01: EMERGENCY STOP

DESCRIPTION

The rover shall provide a physical, latching emergency-stop that removes power from the drive motors when pressed.

RATIONALE

Immediate manual means to stop motion is a baseline safety control.

SOURCE

Safety review; lab policy.

PRIORITY

Critical

VERIFICATION METHOD

Demonstration — press E-stop under motion; motors de-energize.

CONSTRAINTS

E-stop reachable on the chassis top surface.

APPLICABLE STANDARDS

ISO 13850 (emergency stop).

8.2 SA-02: LABORATORY LOTO**DESCRIPTION**

Fabrication equipment shall be used per OSHA lockout/tagout procedures; only the instructor or TAs may remove a lock.

RATIONALE

Prevents injury during fabrication.

SOURCE

CSE Senior Design laboratory policy.

PRIORITY

Critical

VERIFICATION METHOD

Inspection.

CONSTRAINTS

Availability of instructor/TA to remove locks.

APPLICABLE STANDARDS

OSHA 29 CFR 1910.147.

9 SECURITY & PRIVACY REQUIREMENTS**9.1 SE-01: ENCRYPTED CONTROL LINK****DESCRIPTION**

All ground-station↔rover traffic shall be encrypted with TLS 1.3; unauthenticated command sessions shall be refused.

RATIONALE

Prevents hijacking of a moving robot and tampering with reports.

SOURCE

Team; security review.

PRIORITY

Critical

VERIFICATION METHOD

Test — protocol scan; attempt unauthenticated command.

CONSTRAINTS

Pre-shared credential provisioned at setup.

APPLICABLE STANDARDS

IETF RFC 8446 (TLS 1.3).

9.2 SE-02: SITE IMAGERY HANDLING

DESCRIPTION

Captured site imagery shall remain on the rover/ground station and shall not be transmitted off the private network without sponsor approval.

RATIONALE

Solar-site imagery may be sensitive to the sponsor.

SOURCE

Sponsor.

PRIORITY

High

VERIFICATION METHOD

Inspection — data-flow review.

CONSTRAINTS

No cloud egress in prototype.

APPLICABLE STANDARDS

None.

10 MAINTENANCE & SUPPORT REQUIREMENTS

10.1 MS-01: DIAGNOSABLE FAULTS

DESCRIPTION

The rover shall log each subsystem fault with a timestamp and error code retrievable from the ground station without a debugger.

RATIONALE

Field maintainers must localize failures quickly.

SOURCE

Team.

PRIORITY

Moderate

VERIFICATION METHOD

Demonstration.

CONSTRAINTS

Log storage sized for a day of operation.

APPLICABLE STANDARDS

None.

11 OTHER QUALITY ATTRIBUTES

11.1 QA-01: MODULAR PERCEPTION

DESCRIPTION

The fault-detection model shall be replaceable without changes to the navigation or platform software.

RATIONALE

Allows the detector to improve independently as more data is collected.

SOURCE

Team.

PRIORITY

Moderate

VERIFICATION METHOD

Inspection — interface review.

CONSTRAINTS

Defined model interface.

APPLICABLE STANDARDS

ISO/IEC 25010:2023 (modularity) [2].

12 VERIFICATION & TRACEABILITY

Req ID	Requirement (abbrev.)	Method	Acceptance
FR-01	Waypoint patrol within 0.5 m	T	cross-track ≤ 0.5 m
FR-03	On-board fault detection	T	meets PE-03
FR-05	Safe-stop & return	T	all triggers handled
PE-01	≥ 200 panels/hour	T	measured ≥ 200
PE-03	Detection quality	A	recall ≥ 0.80 , prec. ≥ 0.70
SE-01	TLS 1.3 control link	T	scan shows TLS1.3; unauth refused

Full requirement-to-design-to-test traceability is maintained in the team's shared matrix and extended in the ADS, DDS, and Test Plan.

13 FUTURE ITEMS

13.1 FI-01: CLOUD ANALYTICS DASHBOARD

DESCRIPTION

A hosted dashboard aggregating fault trends across runs and sites.

RATIONALE

Adds fleet-level insight but exceeds prototype budget/time.

SOURCE

Sponsor.

PRIORITY

Future

VERIFICATION METHOD

Test (if implemented).

CONSTRAINTS

Requires cloud hosting and data egress.

APPLICABLE STANDARDS

None.

REFERENCES

- [1] ISO/IEC/IEEE, “ISO/IEC/IEEE 29148:2018 — Systems and software engineering — Life cycle processes — Requirements engineering,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2018.
- [2] ISO/IEC, “ISO/IEC 25010:2023 — Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — Product quality model,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2023.